

ReadMe File for Matlab and R Files for “The Minimum Wage in Firms Organizations: Productivity Implications”, *Journal of Political Economy*

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Overview

The code in this replication package contains 4 folders of Matlab code which calibrate and simulate the models used in the paper, and 2 folders of R code which generates a number of the figures in the paper. These code files generate the results found in 8 tables and 12 figures in the paper, and the replicator should expect it to take about 36 hours to run all of the Matlab code if run in parallel, and less than a minute to run the R code.

Data Availability and Provenance Statements

5 of the 6 folders of code do not directly use any data (they do use results from the empirical analysis described in the Stata/SAS section of the replication package). The “R Figures A2-A3” folder uses data that were publicly accessible online. No registration, membership, application procedure or monetary payment was required to obtain the authors’ source files.

Figure A2 – Economic Policy Uncertainty

Data source: Baker, Bloom and Davis (2016) for all countries except Spain; Ghirelli, Perez and Urtasun (2019) for Spain. The data are maintained at Economic Policy Uncertainty (2025).

Source landing page: https://www.policyuncertainty.com/all_country_data.html

Direct workbook URL: https://www.policyuncertainty.com/media/All_Country_Data.xlsx

Authors’ download date recorded in the supplied material: 25 March 2025.

The source page states that the data are licensed under “Creative Commons Attribution 4.0 International (CC BY 4.0)”. The package contains the original XLSX file downloaded from the source page, with a “Read_me” sheet added with the URL and data of download; and the package also contains a non-proprietary CSV export of the numeric observations from the workbook’s “EPU” sheet: “Data/epu_all_country_2025-03-25.csv”. “Code/Graphs_UNCERTAINTY_Bloom_v1.R” selects the six country series used in Figure A2 from the CSV file.

Figure A3 – Kaitz Indices

Data source: Bozio and Wasmer (2024).

Source landing page: <https://www.strategie-plan.gouv.fr/publications/mission-bozio-wasmer-politiques-dexonerations-de-cotisations-sociales-une-inflexion>

Direct Chapter 1 workbook URL: https://www.strategie-plan.gouv.fr/files/files/Publications/Excel/figures_chapitre_1.xlsx

Data of last update of the data: 3 October 2024.

Authors’ access date recorded in the supplied program: 29 August 2026.

The source page states that the data are licensed under “Etalab-2.0”, which permits reuse and redistribution as long as the source and date of last update are provided. The supplied R program uses sheet “Gr10 Kaitz par an et pays”, ranges A6:H25 (labor costs) and J6:Q25 (income). The package contains the original XLSX file downloaded from the source page, as well as the relevant worksheet cells A6:Q25 as a non-proprietary CSV, “Data/kaitz_gr10_source_matrix.csv”; “Code/Graphs_Kaitz_v1.R” performs the exact range selection and reshaping. The source worksheet identifies the underlying source as “OCDE, modèle Tax-Ben”.

Data Rights

The authors have legitimate access to and permission to use the data used for Figures A2 and A3.

- EPU: redistribution is documented by the source’s CC BY 4.0 license.
- Kaitz extract: redistribution is documented by the source’s Etalab-2.0 license
- The new replication code/documentation is licensed under “LICENSE.txt”; third-party data retain their own terms.

Computational Requirements

Software Requirements

- Matlab (code was last run with Matlab release R2024b)
- R (code was last run with R version 4.6.1)

All Matlab code uses base Matlab plus the Global Optimization Toolbox; all R code uses base R, with no external packages used.

Controlled Randomness

- ☐ Random seed is set at line _____ of program _____
- ☒ No Pseudo random generator is used in the analysis described here.

Memory, Runtime, Storage Requirements

Summary Approximate time needed to reproduce the analyses on a standard 2024 desktop machine:

- ☐ <10 minutes
- ☐ 10-60 minutes
- ☐ 1-2 hours
- ☐ 2-8 hours
- ☐ 8-24 hours
- ☒ 1-3 days
- ☐ 3-14 days
- ☐ > 14 days

Approximate storage space needed:

- ☒ < 25 MBytes
- ☐ 25 MB - 250 MB
- ☐ 250 MB - 2 GB
- ☐ 2 GB - 25 GB
- ☐ 25 GB - 250 GB
- ☐ > 250 GB
- ☐ Not feasible to run on a desktop machine, as described below.

Details The code was last run on a **20-core Intel(R) Core(TM) i7-14700K (3.40 GHz)** with **Windows 11 and 32 GB of RAM**. Computation took about **36 hours**.

Description of Programs/Code

- The replication archive contains 4 Matlab folders, 3 of which contain the code needed to generate the results for 3 different specifications of the model that are presented in the paper; the “Sensitivity Specifications” folder is further subdivided into 7 sub-folders, each of which contains the code needed to generate the results for one of the 7 sensitivity analyses considered in Appendix G of the paper. These 10 folders – the 3 folders entitled “Baseline Specification”, “Unemployment Specification”, and “Unproductive Managers Specification”, and the 7 sub-folders in the “Sensitivity Specifications” folder – will be referred to as “analysis folders” in what follows.
 - Each analysis folder is divided into 3 sub-folders: “Code” contains the Matlab code, “Data” is initially empty but will eventually contain the Matlab .mat files that are created by the Matlab code and used in some code files, and “Output” is initially empty but will eventually contain the final output files that are used in the paper. In each analysis folder, there is one master file of Matlab code, entitled “Main.m”, which executes all of the code in that analysis folder. If run on its own, Main.m should take between 12 and 30 hours in each analysis folder (except for the “Unproductive Managers Specification” folder, which only takes about 10 minutes); if the Main.m in all 10 analysis folders are executed simultaneously, the code should take about 36 hours to run. In each Main.m file, lines 5-7 define the paths for the Code, Data, and Output folders, with line 3 defining the base path on the computer we used; replicators will need to replace line 3 in each Main.m file with the appropriate path on their own computer.
- The replication archive also contains 2 R folders, the first of which is entitled “R Simulation Figures”, which is divided into 2 sub-folders: “Code” contains the R scripts, and “Output” is initially empty but will eventually contain the PDF files for a number of figures in the paper. In the Code sub-folder, “Main.R” is the master script which calls all of the other scripts, and should take less than a minute to run; each of the other scripts loads the R packages that are required in that script. Line 2 of Main.R defines the main path for the replication folder, and replicators will need to replace this line with the appropriate path on their own computer.
- The second R folder is entitled “R Figures A2-A3”, which is divided into 3 sub-folders: “Code” contains the R scripts, “Data” contains the 2 original XLSX data files and the 2 analysis CSV data files (as well as a CSV file detailing the third-party rights to the data), and “Output” is initially empty but will eventually contain the PDF files for Figures A2 and A3 in the appendix of the paper. In the Code sub-folder, “Main.R” is the master script which calls all of the other scripts, and should take less than a minute to run; each of the other scripts loads the R packages that are required in that script. Line 2 of Main.R defines the main path for the replication folder, and replicators will need to replace this line with the appropriate path on their own computer.

Instructions to Replicators

- In the “Baseline Specification” folder, run the “Main.m” file in the “Code” sub-folder, updating the path on line 3 to the appropriate folder on your computer. This will run all of the code for the baseline specification of the model, and generate PDFs and CSV files in the “Output” sub-folder for Figures 3, 4, 5, 6, 7, 8, F1, and F2, as well as Tables 5, 6, F1, and F2. For Figures 6, 7, 8, and F2, the CSV files are used to produce the figures using R in the second-last step below.
- In the “Sensitivity Specifications” folder, each sub-folder represents a different sensitivity specification, and thus a different column of Table G1 and a different panel of Figure G1; all sub-folders are labelled with the column number as well as a description of the specification.
 - In each sub-folder, run the “Main.m” file in the “Code” sub-sub-folder, updating the path on line 3 to the appropriate folder on your computer. This will run all of the code for the given sensitivity specification of the model, and generate the CSV file for the given column of Table G1 in the “Output” sub-folder, as well as the CSV file which is used to produce the given panel of Figure G1

using R in the second-last step below.

- In the “Unemployment Specification” folder, run the “Main.m” file in the “Code” sub-folder, updating the path on line 3 to the appropriate folder on your computer. This will run all of the code for the unemployment specification of the model, and generate the CSV files for Table H1 and the upper panel of Table H2 in the “Output” sub-folder, as well as the CSV file which is used to produce the figure in the lower panel of Table H2 using R in the second-last step below.
- In the “Unproductive Managers Specification” folder, run the “Main.m” file in the “Code” sub-folder, updating the path on line 3 to the appropriate folder on your computer. This will run all of the code for the unproductive managers specification of the model, and generate the CSV file for Table I1 in the “Output” sub-folder, as well as the CSV file which is used to produce Figure I1 using R script in the second-last step below.
- In the “R Simulation Figures” folder, run the “Main.R” file in the “Code” sub-folder, updating the working directory on line 2 to the appropriate folder on your computer. This will run all of the code that uses the CSV files produced above to generate PDF files for Figures 6, 7, 8, F2, G1, and I1, and the figure in the lower panel of Table H2.
- In the “R Figures A2-A3” folder, run the “Main.R” file in the “Code” sub-folder, updating the working directory on line 2 to the appropriate folder on your computer. This will run all of the code that uses the data in the “Data” sub-folder to generate PDF files for Figures A2 and A3.

List of Tables and Programs

The provided code reproduces:

- ☐ All numbers provided in text in the paper
- ☐ All tables and figures in the paper
- ☒ Selected tables and figures in the paper, as explained and justified below.

Table or Figure #	Program	Line Number	Note
Table 5	Baseline Specification/Code/Calibration.m	427-429	
Table 6	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	496-509	
Table F1	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	511-526	
Table F2	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	528-542	
Table G1	Sensitivity Specifications/Sensitivity (i) - 1+ Only/Code/Calibration.m	421-425	Column (i)
	Sensitivity Specifications/Sensitivity (ii) - GM6 & 1+/Code/Calibration.m	421-425	Column (ii)
	Sensitivity Specifications/Sensitivity (iii) - SelfEmployment, 1+ Only/Code/Calibration.m	424-428	Column (iii)
	Sensitivity Specifications/Sensitivity (iv) - SelfEmployment, GMR6 & 1+/Code/Calibration.m	424-428	Column (iv)
	Sensitivity Specifications/Sensitivity (v) - Hours Calibration/Code/Calibration.m	425-429	Column (v)
	Sensitivity Specifications/Sensitivity (vi) - Kaitz Calibration/Code/Calibration.m	422-427	Column (vi)
	Sensitivity Specifications/Sensitivity (vii) - Mean Ratio Calibration/Code/Calibration.m	422-427	Column (vii)
Table H1	Unemployment Specification/Code/Calibration.m	439-443	

Table or Figure #	Program	Line Number	Note
Table H2	Unemployment Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	410-421 383-407	Upper panel CSV file (lower panel)
	R Simulation Figures/Code/TableH2.R	143-157	PDF file (lower panel)
Table I1	Unproductive Managers Specification/Code/Calibration.m	140-143	
Figure 3	Baseline Specification/Code/CR-HFigures/ CostMinimizationMW.m	121-131 108-118	Panel A Panel B
Figure 4	Baseline Specification/Code/CR-HFigures/ CostMinimizationMW_h.m	91-101	Panel A
	Baseline Specification/Code/CR-HFigures/ CostMinimizationMW_c.m	92-102	Panel B
	Baseline Specification/Code/CR-HFigures/ CostMinimizationMW_A.m	91-101	Panel C
	Baseline Specification/Code/CR-HFigures/ CostMinimizationMW_lambda.m	91-101	Panel D
Figure 5	Baseline Specification/Code/CR-HFigures/ CostMinimizationMW_mw.m	90-100 103-113	Panel A Panel B
Figure 6	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	447-475	CSV file
	R Simulation Figures/Code/Figure_6_legFigG1.R	135-140 151-156	PDF file (legend) PDF file (figure)
Figure 7	Baseline Specification/Code/Decomposition/ ProfitMaximizationMW_GE.m	133-146	CSV file
	R Simulation Figures/Code/Figure_7.R	88-108	PDF files
Figure 8	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	478-487	CSV file
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output32/H_Small/ProfitMaximizationMW_GE.m	327-336	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output32/C_Small/ProfitMaximizationMW_GE.m	328-337	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output32/Lambda_Large/ProfitMaximizationMW_GE.m	327-336	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output32/A_Large/ProfitMaximizationMW_GE.m	327-336	
	R Simulation Figures/Code/Figure_8.R	149-203	PDF files
Figure A2	R Figures A2-A3/Code/Graphs_UNCERTAINTY_Bloom_v1.R	37	PDF file
Figure A3	R Figures A2-A3/Code/Graphs_Kaitz_v1.R	49 86	PDF file (Panel A) PDF file (Panel B)
Figure F1	Baseline Specification/Code/AlphaFigures/ ProfitMaximizationMW_GE.m	123-138 140-159 161-176 178-197	Panel A Panel B Panel C Panel D

Table or Figure #	Program	Line Number	Note
Figure F2	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	478-487	CSV file (Panel A)
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output32/H_Small/ProfitMaximizationMW_GE.m	327-336	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output32/C_Small/ProfitMaximizationMW_GE.m	328-337	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output32/Lambda_Large/ProfitMaximizationMW_GE.m	327-336	CSV file (Panel B)
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output32/A_Large/ProfitMaximizationMW_GE.m	327-336	
	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	490	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output64/H_Small/ProfitMaximizationMW_GE.m	327-336	CSV file (Panel C)
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output64/C_Small/ProfitMaximizationMW_GE.m	328-337	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output64/Lambda_Large/ProfitMaximizationMW_GE.m	327-336	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output64/A_Large/ProfitMaximizationMW_GE.m	327-336	CSV file (Panel C)
	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	493	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output96/H_Small/ProfitMaximizationMW_GE.m	327-336	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output96/C_Small/ProfitMaximizationMW_GE.m	328-337	PDF files
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output96/Lambda_Large/ProfitMaximizationMW_GE.m	327-336	
	Baseline Specification/Code/MWResults/MWResults_ Technology_Output96/A_Large/ProfitMaximizationMW_GE.m	327-336	
Figure G1	R Simulation Figures/Code/Figure_F2.R	111-184	PDF files
	Sensitivity Specifications/Sensitivity (i) - 1+ Only/Code/MWResults/ProfitMaximizationMW_GE.m	445-473	CSV file (Panel i)
	Sensitivity Specifications/Sensitivity (ii) - GM6 & 1+/Code/MWResults/ProfitMaximizationMW_GE.m	445-473	CSV file (Panel ii)
	Sensitivity Specifications/Sensitivity (iii) - SelfEmployment, 1+ Only/Code/MWResults/ProfitMaximizationMW_GE.m	454-482	CSV file (Panel iii)
	Sensitivity Specifications/Sensitivity (iv) - SelfEmployment, GMR6 & 1+/Code/MWResults/ProfitMaximizationMW_GE.m	454-482	CSV file (Panel iv)
	Sensitivity Specifications/Sensitivity (v) - Hours Calibration/Code/MWResults/ProfitMaximizationMW_GE.m	445-473	CSV file (Panel v)
	Sensitivity Specifications/Sensitivity (vi) - Kaitz Calibration/Code/MWResults/ProfitMaximizationMW_GE.m	445-473	CSV file (Panel vi)
	Sensitivity Specifications/Sensitivity (vii) - Mean Ratio Calibration/Code/MWResults/ProfitMaximizationMW_GE.m	445-473	CSV file (Panel vii)
	R Simulation Figures/Code/Figure_G1.R	114-147	PDF files (figures)
	R Simulation Figures/Code/Figure_6_legFigG1.R	142-147	PDF files (legend)

Table or Figure #	Program	Line Number	Note
Figure I1	Unproductive Managers Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	96-109	CSV file (Panel A)
	Baseline Specification/Code/MWResults/ ProfitMaximizationMW_GE.m	447-475	CSV file (Panel B)
	R Simulation Figures/Code/Figure_I1.R	152-183	PDF files

References

Baker, Scott R., Nicholas Bloom, and Steven J. Davis (2016): “Measuring Economic Policy Uncertainty,” *Quarterly Journal of Economics* 131(4): 1593–1636.

Bozio, Antoine, and Etienne Wasmer (2024): “Les politiques d’exonérations de cotisations sociales : une inflexion nécessaire,” Mission Bozio-Wasmer, France Strategie/Haut-commissariat à la stratégie et au plan. Data portal: <https://www.strategie-plan.gouv.fr/publications/mission-bozio-wasmer-politiques-dexonérations-de-cotisations-sociales-une-inflexion>.

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